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A Motion Decoupling Control Based on Differential Geometry for Distributed Drive Articulated Heavy Vehicle

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Abstract
A vehicle system motion decoupling control method was proposed to address challenges in controlling articulated heavy vehicles (AHVs). The method, based on differential geometry theory, focused on distributed electric drive AHVs. Its objective was to separate the highly nonlinear and strongly coupled dynamics system into two relatively independent subsystems: longitudinal and lateral motions. Additionally, a robust controller was designed to improve the vehicle's resistance to external disturbances like side winds. Simulation tests using a TruckSim model of a distributed electric drive AHV show significant improvements compared to vehicles without decoupling control. The rearward amplification (RA) is reduced by 4.5%, the longitudinal velocity deviation by 67.5%, and the yaw rate deviation by 69.7%. The vehicle also demonstrates enhanced stability when subjected to strong breeze disturbances. To validate the control performance in real-time systems, the hardware-in-the-loop tests were conducted, which confirms the effectiveness of the proposed control approach in practical applications.

Keywords Distributed drive · Differential geometry · Decoupling control · Robust control

List of symbols

a_1	Distance from tractor's front axle to its center of gravity, m
a_2	Distance from trailer's center of gravity to hinge point, m
b_1	Distance from tractor's rear axle to its center of gravity, m
b_2	Distance from trailer's axle to its center of gravity, m
c	Distance from tractor's center of gravity to hinge point, m
F_{Hx}	Longitudinal force applied at hinge point, N
F_{Hy}	Lateral force applied at hinge point, N
F_{yf}	Lateral force exerted by tractor's front axle tires, N
F_{yr}	Lateral force exerted by tractor's rear axle tires, N
F_{ys}	Lateral force exerted by trailer's axle tires, N
I_{z1}	Yaw inertia of tractor, kg m^2
I_{z2}	Yaw inertia of trailer, kg m^2
i	Steering ratio
K	Stability factor of vehicle
k_1	Cornering stiffness of tractor's front axle, N/rad
k_2	Cornering stiffness of tractor's rear axle, N/rad
m_1	Mass of tractor, kg
m_2	Mass of trailer, kg
M_{z1}	Active yaw moment of tractor, N m
M_{z2}	Active yaw moment of trailer, N m
m_1	Mass of tractor, kg
m_2	Mass of trailer, kg
R_1	Radius of tractor tires, m
R_2	Radius of trailer tires, m
T_{l1}	Driving moment of tractor's left wheels, N m
T_{r1}	Driving moment of tractor's right wheels, N m
T_{l2}	Driving moment of trailer's left wheels, N m
T_{r2}	Driving moment of trailer's right wheels, N m
T_{l1}^*	Desired driving moment of tractor's left wheels, N m
T_{r1}^*	Desired driving moment of tractor's right wheels, N m
T_{l2}^*	Desired driving moment of trailer's left wheels, N m
T_{r2}^*	Desired driving moment of trailer's right wheels, N m
v_{x1}	Longitudinal velocity of the tractor, m/s
v_{x2}	Longitudinal velocity of the trailer, m/s
y_1	Lateral displacement of tractor's center of mass, m
y_2	Lateral displacement of trailer's center of mass, m

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α_{pedal} Accelerator pedal position, [0–1]
β_1 Slip angle of tractor, rad
β_2 Slip angle of trailer, rad
δ Front axle steering angle, rad
δ_{sw} Steering wheel steering angle, rad
θ Hinge angle, rad
ψ_1 Yaw angle of tractor, rad
ψ_2 Yaw angle of trailer, rad
ω_1 Yaw rate of tractor, rad/s
ω_2 Yaw rate of trailer, rad/s

Subscripts

ref Reference value
max Maximum value

1 Introduction

In the realm of vehicle dynamics, the distributed electric drive articulated heavy vehicle (AHV) is equipped with propulsion motors positioned on each wheel, offering precise control over the output torque of all drive motors for longitudinal force during forward motion. Additionally, this configuration enables the adjustment of output torque in select drive motors to provide active yaw-moment assistance for steering. While this distributed drive system enhances stability control, it also introduces complexities in power coupling within the vehicle, making system control challenging. Particularly, during medium- and high-speed conditions, the AHV behaves as a complex multivariable system with interdependencies among longitudinal, lateral, and yaw motions, influenced by steering wheel angle. The coupling between the drive/brake and steering subsystems creates additional yaw moments during operation, making the control of yaw stability a critical concern. Numerous control algorithms exist for the management of vehicular lateral stability, each possessing its distinctive advantages. Feed-forward control, for instance, adeptly anticipates vehicular behavior in accordance with the vehicle's dynamics, thus enabling the attainment of precision-driven stability control and the mitigation of instability risks. Conversely, feedback control exhibits the ability to dynamically tailor its control strategy in response to the present vehicle state, thereby accommodating diverse driving scenarios and varying road conditions (Kati et al., 2021). Linear quadratic regulator (LQR) control, grounded in a linear dynamics model, distinguishes itself through its mathematical simplicity and ease of implementation, facilitating seamless debugging (Chen et al., 2023; Han et al., 2018). Model predictive control (MPC), in contrast, excels in prognosticating forthcoming vehicle behavior over a specified time horizon, thus enhancing its capacity to effectively navigate evolving road conditions and driving exigencies (Chen & Yao, 2023; Liu et al., 2022). Furthermore, the integration of intelligent control techniques in conjunction with machine learning imparts the ability to discern intricate nonlinear relationships, enabling adaptability to diverse driving situations and bolstering control performance (Wang et al., 2022). Nonetheless, these aforementioned control algorithms tend to overlook unwanted oscillations in the longitudinal motion of the vehicle, stemming from additional control actions undertaken to ameliorate lateral stability. The considerable mass, the elevated center of gravity, and the interaction between the trailer and tractor make the AHV susceptible to yaw-induced stability issues, including yawing, rollovers, and significant off-road accidents (b; Deng et al., 2022a; Trigell et al., 2017). Addressing this challenge and achieving precise stability control while distributing wheel drive torque effectively necessitate decoupling or approximating the coupling between longitudinal and lateral motions in the AHV.

Numerous researchers have extensively investigated the decoupling problem in vehicle kinematics and systems. Studies have explored the impact of magnetic imbalance resulting from deformation in the air gap of hub motors on coupled dynamics during vehicle motion (Tan & Lu, 2016). Platoon driving with identical controllers and rigid information flow has been examined, along with the challenges posed by weak wireless communication, which was addressed through decoupling distributed H control for vehicular platoons (Gao et al., 2019). Additionally, approximately decoupled control laws for longitudinal, lateral, and yaw motions have been developed, achieving the desired decoupling effect with a new input–output mapping relationship (Chen et al., 2012), and a nonlinear decoupling control method was presented for a four-wheel steering (4WS) vehicle (Chen & Jia, 2012).

In the context of active steering control techniques, gain-scheduled steering controllers have been developed to achieve lateral-yaw decoupling and yaw rate damping based on feedback of yaw rate and front steering angle, with adjustable gains depending on vehicle speed (Zheng & Anwar, 2009). Active 4WS control using proportional-integral (PI) control has also been employed to decouple lateral velocity and sideslip angle dynamics (Marino & Scalzi, 2010). Furthermore, vehicle kinematic control strategies have been proposed to enable decoupling of front and rear-wheel active steering, utilizing sliding mode control theory (Hiraoka et al., 2009). A three-preview-point driver model and LQR active trailer steering (ATS) strategy have been proposed to decouple lateral and yaw motions, which improved both maneuverability and stability of the AHV (Deng et al., 2023).

In the field of direct yaw-moment control (DYC) technology, antilock braking systems (ABSs) with nonlinear sliding mode control methods (SMC) have been implemented for uniform distribution of longitudinal tire-road friction forces, decoupling various interactions of the vehicle (Rajendran et al., 2019). Theoretical computational methods have been introduced to decouple dynamic motions, enabling independent control of roll, pitch, and yaw motions through drive force distribution control on each wheel (Katsuyama, 2013). Integrated control strategies using nonlinear sliding mode control algorithms have been presented for decoupling control and allocation of tire forces in the longitudinal, lateral, and vertical directions (Wang et al., 2023; Zhao et al., 2019). Torque vectoring control strategies have incorporated two-level allocation formulas to achieve decoupling of the torque vector control between the front/rear axle and the left/right wheels (Hu et al., 2020). Moreover, linear parameter varying (LPV) control and H_∞ robust control approaches have been proposed to achieve decoupling performance during vehicle steering at different speeds (Li & Jia, 2014, 2016; Li et al., 2014).

As research progressed, the complexity of vehicle models evolved from decoupled control of vehicle dynamics through two subsystems to decoupled control of multiple subsystems. Coordination control technologies for vehicle multi-subsystems have been developed, involving motion decoupling control methods and adaptive model predictive control for high-speed vehicle conditions (Liang et al., 2019, 2021). Nonlinear decoupling controllers have been designed to separate the vehicle system into three second-order subsystems for longitudinal acceleration/brake force control velocity, front-wheel steering control for lateral motion, and rear-wheel steering control for lateral-yaw motion (Jia, 2000). Other compensatory robust decoupling control systems have been proposed, incorporating robust decoupling controllers (RDC) and feedback compensation controllers (FCC) for stability control in distributed driving electric vehicles (DDEV) through active front steering (AFS) and wheel driving force allocation (Shi et al., 2019). Strategies based on differential geometry have been employed to decouple vehicle active steering and active suspension, as well as braking and active suspension (Chen, 2013). A neural network inverse system and a proportional-integral–differential (PID) feedback control strategy have been combined to tackle the yaw and rolling motions decoupling problem in multi-input multi-output (MIMO) nonlinear systems, including suspension systems coupled with other subsystems (Xu et al., 2022). However, the utilization of motion decoupling control based on differential geometry presents two challenging aspects that warrant careful consideration: (1) the depiction of the objective reality encounters intricacies when seeking a precise mathematical portrayal. Foremost, the establishment of a coherent mathematical model for the subject under investigation becomes imperative. This involves deriving unambiguous mathematical formulations, which constitute a prerequisite for the effective employment of differential geometry in addressing nonlinear predicaments. As such, a judicious simplification of the intricate vehicle model becomes necessary; (2) an accurate determination of the system's relative order assumes paramount significance within this methodology. An erroneous assessment of the relative order could induce inaccuracies in the system's decoupling matrix, thereby impeding the achievement of the intended control objectives. Therefore, decoupling control may benefit from combination with other control algorithms to enhance anti-disturbance performance. For instance, control schemes based on neural network inverse system decoupling have been developed for traditional vehicle active suspension and active steering systems, strengthened by the inclusion of internal model controllers and robust controllers to handle unknown disturbances (Wang et al., 2018).

In addition, the accuracy and real-time performance of the acquisition of vehicle motion parameters are quite important for the performance of the decoupled control. The existing approaches for acquiring certain vehicular motion parameters (such as steering wheel angle, vehicle speed, lateral acceleration, sideslip angle, and yaw rate) through sensor measurements and algorithmic estimations have reached a higher level of maturity (Anderson & Bevly, 2010, b; Deng et al., 2022a; Kim et al., 2015). Somewhat distinct from conventional passenger cars, the precise determination of the hinge angle holds paramount significance for AHVs, along with other articulated vehicles. Excessive hinge angles in these cases can potentially result in catastrophic jackknifing of the articulated vehicle (Trigell et al., 2017). Given the heightened emphasis on enhancing the driving safety and intelligence of articulated vehicles, certain scholars have directed their attention toward the quantification and estimation of the hinge angle (Bahramgiri et al., 2023; Korayem et al., 2021). This study rests upon the supposition that the tractor's longitudinal speed, sideslip angle, yaw rate, front axle steering angle, and the AHV's hinge angle can be ascertained through measurement or estimation.

This study focuses on enhancing the effectiveness of decoupling control in distributed AHVs and improving their resilience. Specifically, we examine a front-wheel-steering, distributed electric drive AHV and develop a three-degree-of-freedom vehicle model encompassing longitudinal, lateral, and yaw motions. Employing the differential geometry theory, we aim to decouple the longitudinal and yaw motions of the vehicle and devise a decoupling control strategy in conjunction with robust control theory. To validate our approach, we conduct co-simulations under multiple operating conditions using TruckSim/Simulink and corroborate the results through hardware-in-the-loop simulation (HILs) data.

2 Vehicle Dynamics Model

2.1 Distributed Electric Drive AHV Model

Presented in Fig. 1, the simplified model of an AHV serves as a prototype, while Table 1 provides the corresponding vehicle parameters. Under a certain condition, the discrepancy between the left front-wheel steering angle δ_l and the right front-wheel steering angle δ_r is negligible. Consequently, it can be simplified as the front wheels steering angle δ, where $\delta = \delta_l = \delta_r$. Furthermore, sin$\delta$ can be approximated as δ, and cosδ can be approximated as 1. For the sake of simplicity, this model does not account for vertical, pitching, and rolling motions, while the tire cornering characteristics are assumed to operate within the linear range. Consequently, a three-degree-of-freedom nonlinear vehicle dynamics model is established, encompassing longitudinal, lateral, and yaw motions. The kinematic relationships in each direction are described as follows.

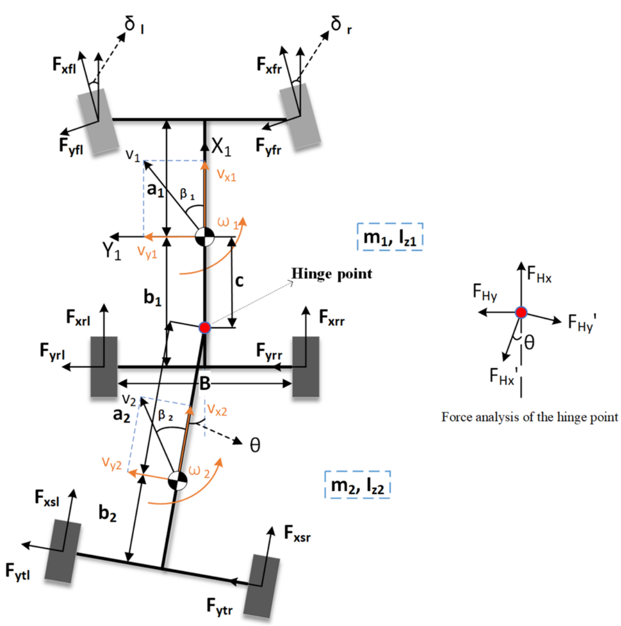

Fig. 1 Three-degree-of-freedom vehicle model

The expressions for the longitudinal motion are given in Eqs. (1) and (2)

$$m_1\dot{v}_{x1} = \frac{(T_{l1}+T_{r1})}{R_1} + v_{x1}\beta_1\omega_1 - F_{yf}\delta - F_{Hx}\Delta F_{x1} \tag{1}$$

$$m_2\dot{v}_{x2} = \frac{(T_{l2}+T_{r2})}{R_2} + v_{x2}\beta_2\omega_2 + F_{ys}\theta + F_{Hx}\Delta F_{x2}, \tag{2}$$

where F_{Hx} denotes the longitudinal force of the hinge point, while v_{x1} and v_{x2} represent the longitudinal velocities of the tractor and trailer, respectively. Additionally, β_1 and β_2 correspond to the slip angles of the tractor and trailer, respectively. The variables ω_1 and ω_2 represent the yaw rates of the tractor and trailer, respectively. The lateral force exerted by the tractor's front axle tires is denoted as F_{yf}, where $F_{yf} = F_{yfr} + F_{yfl}$, and the lateral force on the trailer's axle tires is denoted as F_{ys}, with $F_{ys} = F_{ysl} + F_{ysr}$. The symbol θ represents the hinge angle. Moreover, T_{l1} and T_{r1} represent the driving moments of the tractor's left and right wheels, respectively, while T_{l2} and T_{r2} denote the driving moments of the trailer's left and right wheels, respectively. Finally, ΔF_{x1} represents the longitudinal force of the tractor subject to external uncertainties, and ΔF_{x2} represents the longitudinal force of the trailer subject to external uncertainties.

As θ is considered to be a small value, it is reasonable to assume that $v_{x2} = (1-\theta)v_{x1}$. Additionally, by combining Eq. (1) with Eq. (2), the intermediate quantity F_{Hx} can be utilized to derive Eq. (3)

$$\begin{aligned}\big(m_1 + m_2(1-\theta)\big)\dot{v}_{x1} = {} & 2v_{x1}\beta_1\omega_1 - F_{yf}\delta \\ & + m_2\big(1-\dot{\theta}\big)v_{x1} \\ & + v_{x2}\beta_2\dot{\theta} + F_{ys}\theta \\ & + \frac{T_1}{R_1} + \frac{T_2}{R_2} + \Delta F_x.\end{aligned} \tag{3}$$

Table 1 Main vehicle parameters

Parameter	Description	Unit	Value
m_1	Mass of tractor	kg	6310
m_2	Mass of trailer	kg	20,925
I_{z1}	Yaw inertia of tractor	kg m^2	19,665
I_{z2}	Yaw inertia of trailer	kg m^2	439,992
R_1	Radius of tractor tires	M	0.51
R_2	Radius of trailer tires	m	0.522
a_1	Distance from tractor's front axle to its center of gravity	m	1.385
a_2	Distance from trailer's center of gravity to hinge point	m	2.812
b_1	Distance from tractor's rear axle to its center of gravity	m	4.25
b_2	Distance from trailer's axle to its center of gravity	m	4.308
c	Distance from tractor's center of gravity to hinge point	m	4.25

In the given equations, the total driving moment of the tractor is represented as $T_1 = T_{l1} + T_{r1}$, while the total driving moment of the trailer is denoted as $T_2 = T_{l2} + T_{r2}$. Furthermore, $\Delta F_x = \Delta F_{x1} - \Delta F_{x2}$, and $\dot{\theta} = \omega_2 - \omega_1$, where $\dot{\theta}$ signifies the hinge angular velocity.

The expressions for the lateral motion of the vehicle are given in Eqs. (4), (5), (6), and (7)

$$m_1\ddot{y}_1 = F_{yf} + F_{yr} - F_{Hy} \tag{4}$$

$$m_2\ddot{y}_2 = F_{ys} - F_{xs}\theta + F_{Hy} \tag{5}$$

$$\ddot{y}_1 = \dot{v}_{y1} + v_{x1}\omega_1 \tag{6}$$

$$y_2 = y_1 - c\psi_1 - a_2\psi_2, \tag{7}$$

where y_1 and y_2 represent the lateral displacements of the center of mass for the tractor and trailer, respectively. Similarly, ψ_1 and ψ_2 denote the yaw angles of the tractor and trailer, respectively. F_{Hy} corresponds to the lateral force applied at the hinge point. The lateral force exerted by the rear axle tires of the tractor is denoted as F_{yr}, which can be further expressed as $F_{yr} = F_{yrr} + F_{yrl}$. By combining Eqs. (4), (5), (6), and (7), Eq. (8) is obtained.

$$\begin{aligned}\dot{v}_{y1} = {} & \frac{m_2(c+a_2)}{m_1+m_2}\dot{\omega}_1 + \frac{m_2a_2}{m_1+m_2}\ddot{\theta} \\ & + \frac{F_{yf} + F_{yr} + F_{ys} - \frac{T_{l2}+T_{r2}}{R_2}\theta}{m_1+m_2} - v_{x1}\omega_1.\end{aligned} \tag{8}$$

The yaw motion of the tractor is shown in Eq. (9)

$$I_{z1}\dot{\omega}_1 = a_1F_{yf} - b_1F_{yr} - cF_{Hy} + M_{z1} + \Delta M_{z1}. \tag{9}$$

By combining Eqs. (5), (6), (8), and (9), obtain the following equation:

$$\begin{aligned}C_1\dot{\omega}_1 + C_2\ddot{\theta} = {} & \left(\frac{m_2c}{m_1+m_2} + a_1\right)F_{yf} \\ & + \left(\frac{m_2c}{m_1+m_2} - b_1\right)F_{yr} \\ & + \left(\frac{m_2c}{m_1+m_2} - c\right)F_{ys} \\ & + \left(c - \frac{m_2c}{m_1+m_2}\right)\frac{T_2}{R_2}\theta \\ & + M_{z1} + \Delta M_{z1},\end{aligned} \tag{10}$$

where $C_1 = \frac{m_1m_2c(c+a_2)}{m_1+m_2} + I_{z1}$, $C_2 = \frac{m_1m_2ca_2}{m_1+m_2}$, M_{z1} is the active yaw moment of the tractor and ΔM_{z1} is the yaw moment of the tractor disturbed by external uncertainties.

The yaw motion of the trailer is shown in Eq. (11):

$$I_{z2}\dot{\omega}_2 = -b_2F_{xs}\theta - b_2F_{ys} + a_2F_{Hy} + M_{z2} + \Delta M_{z2} \tag{11}$$

By combining Eqs. (4), (6), (8), and (11), obtain the following equation:

$$\begin{aligned}A_1\dot{\omega}_1 + A_2\ddot{\theta} = {} & \frac{m_2a_2}{m_1+m_2}F_{yf} + \frac{m_2a_2}{m_1+m_2}F_{yr} \\ & + \left(\frac{m_2a_2}{m_1+m_2} - a_2 - b_2\right)F_{ys} \\ & + \left(a_2 + b_2 - \frac{m_2a_2}{m_1+m_2}\right)\frac{T_2}{R_2}\theta \\ & + M_{z2} + \Delta M_{z2}\end{aligned} \tag{12}$$

where $A_1 = \frac{m_1m_2a_2(c+a_2)}{m_1+m_2} + I_{z2}$, $A_2 = \frac{m_1m_2a_2^2}{m_1+m_2} + I_{z2}$, M_{z2} is the active yaw moment of the trailer, and ΔM_{z2} is the yaw moment of the trailer disturbed by external uncertainties.

The state space equation is formulated by combining Eqs. (3), (10) and (12), where the state vector $\boldsymbol{x} = \begin{bmatrix} v_{x1} \\ \omega_1 \\ \dot{\theta} \end{bmatrix}$ and the input vector $u = \begin{bmatrix} T_1 \\ M_{z1} \\ M_{z2} \end{bmatrix}$. Since the linear tire model is insufficient in accurately representing the tires' stress state due to significant variations in tire load, a nonlinear tire model is developed utilizing the look-up table method based on the tires' corresponding parameters provided in TruckSim. To ensure that the coefficient matrix of the input vector is of full rank, T_1 is employed as the vehicle state variable when deriving the differential geometric decoupling control law. The state space equation is depicted in Eq. (13)

$$\begin{aligned}\mathrm{M}\begin{bmatrix} \dot{v}_{x1} \\ \dot{\omega}_1 \\ \ddot{\theta} \end{bmatrix} = \begin{bmatrix} f_1 \\ f_2 \\ f_3 \end{bmatrix} + \begin{bmatrix} \frac{1}{R_1} & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}\begin{bmatrix} T_1 \\ M_{z1} \\ M_{z2} \end{bmatrix} + \\ \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}\begin{bmatrix} \Delta F_x \\ \Delta M_{z1} \\ \Delta M_{z2} \end{bmatrix}\end{aligned} \tag{13}$$

where

$$\begin{aligned}f_1 = {} & 2v_{x1}\beta_1\omega_1 + m_2(1-\dot{\theta})v_{x1} + v_{x1}\beta_1\dot{\theta} - F_{yf}\delta \\ & + F_{ys}\theta + \frac{T_2}{R_2}\end{aligned},$$

$$\begin{aligned}f_2 = {} & \left(\frac{m_2c}{m_1+m_2} + a_1\right)F_{yf} + \left(\frac{m_2c}{m_1+m_2} - b_1\right)F_{yr} \\ & + \left(\frac{m_2c}{m_1+m_2} - c\right)F_{ys} + \left(c - \frac{m_2c}{m_1+m_2}\right)\frac{T_2}{R_2}\theta\end{aligned},$$

$$f_3 = \frac{m_2 a_2}{m_1 + m_2} F_{yf} + \frac{m_2 a_2}{m_1 + m_2} F_{yr} + \left(\frac{m_2 a_2}{m_1 + m_2} - a_2 - b_2 \right) F_{ys} \ , + \left(a_2 + b_2 - \frac{m_2 a_2}{m_1 + m_2} \right) \frac{T_2}{R_2} \theta$$

$$\mathrm{M} = \begin{bmatrix} m_1 + m_2(1-\theta) & 0 & 0 \\ 0 & \frac{m_1 m_2 c(c+a_2)}{m_1+m_2} + I_{z1} & \frac{m_1 m_2 c a_2}{m_1+m_2} \\ 0 & \frac{m_1 m_2 a_2 (c+a_2)}{m_1+m_2} + I_{z2} & \frac{m_1 m_2 a_2^2}{m_1+m_2} + I_{z2} \end{bmatrix},$$

where the system input vector $u = \begin{bmatrix} T_1 \\ M_{z1} \\ M_{z2} \end{bmatrix}$ represents a non-diagonal matrix of coefficients. Consequently, variations in M_{z1} and M_{z2} due to yaw motion demands lead to changes in the longitudinal motion state variable v_x. As a result, the longitudinal and lateral controls interact and mutually influence each other. The nonlinear vehicle exhibits a coupling relationship between the longitudinal and yaw motions, necessitating the implementation of decoupling control.

2.2 Maneuvering Signals

The driver's inputs for maneuvering the vehicle consist of the accelerator pedal position α_{pedal} and the steering wheel steering angle δ_{sw}. These inputs correspond to the reference longitudinal velocity of the tractor $v_{x1\mathrm{ref}}$ and the reference steering radius of the tractor $SR_{1\mathrm{ref}}$, respectively.

This is an example of the equation.

A linear relationship can be established between the reference longitudinal velocity of the tractor and the accelerator pedal position. Specifically, the reference longitudinal velocity of the tractor $v_{x1\mathrm{ref}} = \alpha_{\mathrm{pedal}} v_{\max}$, where $v_{\max}$ is the maximum velocity.

In the steady-state response of the vehicle, where $\beta_{1\mathrm{ref}} = 0$ and $\omega_{1\mathrm{ref}} = \omega_{2\mathrm{ref}}$, the reference steering radius of the tractor (Yin & Jiang, 2019) is given in Eq. (14)

$$SR_{1\mathrm{ref}} = \frac{\left(1 + K v_{x1\mathrm{ref}}^2\right) l_1}{i \delta_{sw}}; \tag{14}$$

Meanwhile, the reference yaw rate of the tractor is shown in Eq. (15)

$$\omega_{1\mathrm{ref}} = \frac{v_{x1\mathrm{ref}}}{SR_{1\mathrm{ref}}}; \tag{15}$$

combining Eqs. (14) and (15) yields Eq. (16)

$$\omega_{1\mathrm{ref}} = \omega_{2\mathrm{ref}} = \frac{v_{x1\mathrm{ref}} \delta}{\left(1 + K v_{x1\mathrm{ref}}^2\right) l_1}, \tag{16}$$

where $\delta = i \delta_{sw}$, i is the steering ratio; K is the stability factor, see Eq. (17) (Bai et al., 2020)

$$K = \frac{1}{l_1^2 l_2} \left(\frac{a_1 l_1 m_1 + (a_1 + c) b_2 m_2}{k_2} - \frac{b_1 l_2 m_1 + (b_1 - c) b_2 m_2}{k_1} \right), \tag{17}$$

where $l_1 = a_1 + b_1$, $l_2 = a_2 + b_2$, and k_1 and k_2 are the lateral deflection stiffnesses of the tractor's front and rear axles, respectively, calculated by the tire model based on the variation of the vertical load.

3 Control System Design

The driving control of the AHV necessitates accurate responses to the driver's maneuvering signals, which entails precise tracking of the reference vehicle velocity and yaw rate, while maintaining high resilience against external disturbances like wind interference. Differential geometric decoupling, through feedback linearization, can transform a nonlinear system into multiple univariate pseudo-linear systems, significantly simplifying control and reducing mutual perturbations between subsystems. However, differential geometric decoupling demands high model accuracy and is less tolerant to disturbances such as model-based jitter. In contrast, robust control employs closed-loop feedback to handle uncertainties, enabling better control quality for linear systems in the presence of disturbances and enhancing the immunity of the differential geometric decoupling control. Consequently, the design of the overall vehicle control system can be divided into two modules: the vehicle system decoupling controller and the robust controller.

The operational procedures of the proposed robust controller and decoupling controller are as follows: (1) Computation of the error gain: The robust controller receives the real-time motion parameters of the AHV, encompassing the longitudinal velocity of the tractor v_{x1}, the tractor's yaw rate ω_1 and the hinge angle θ. These parameters are combined with the AHV's mechanical characteristics for the real-time determination of the error gain. This process is executed employing the linear matrix inequality (LMI) toolbox within MATLAB; (2) Computation of the virtual input $\boldsymbol{v}$ for the decoupling controller: The decoupling controller derives the virtual input by calculating the deviation between the real-time motion parameters of the AHV and the reference motion parameters of the AHV ($v_{x1\mathrm{ref}}$, $\omega_{1\mathrm{ref}}$, θ_{ref}) obtained from the reference model. This calculation yields the error $\boldsymbol{e}$ pertaining to the vehicle's motion

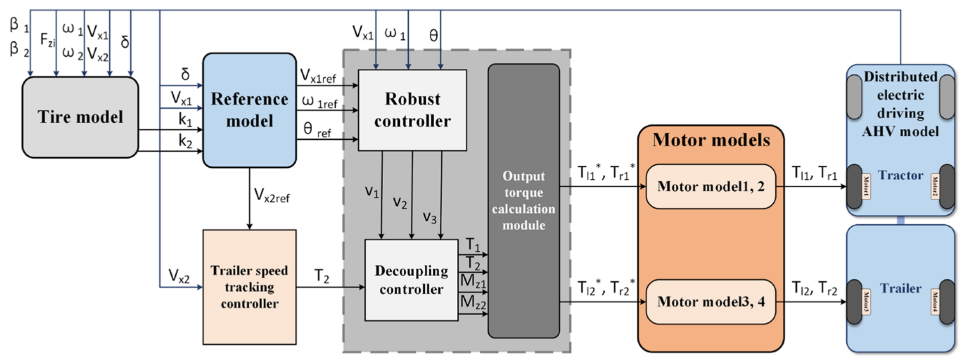

Fig. 2 Structure of the proposed vehicle control system

parameters. Subsequently, the robust controller multiplies this error $\boldsymbol{e}$ by the previously computed error gain, thereby yielding the output $\boldsymbol{v}$, which serves as the virtual input for the decoupling controller; (3) Calculation of the required driving torque and yaw torque: The decoupling controller is responsible for determining the required torque for the tractor T_1 and trailer T_2, as well as the tractor's yaw moment M_1 and the trailer's yaw moment M_2. This computation is achieved through the decoupling process, yielding the aforementioned outputs. The structure of the proposed vehicle control system is depicted in Fig. 2.

3.1 Differential Geometry Decoupling Controller Design

As shown in Eq. (18), the original system can be written as a typical affine nonlinear system structure

$$\begin{cases} \dot{\boldsymbol{x}} = f(\boldsymbol{x}) + \mathrm{g}_1(\boldsymbol{x})u_1 + \mathrm{g}_2(\boldsymbol{x})u_2 + \mathrm{g}_3(\boldsymbol{x})u_3 \\ \quad + \quad p(\boldsymbol{x})\boldsymbol{w} \\ \boldsymbol{y} = \quad h(\boldsymbol{x}) \end{cases}, \tag{18}$$

where $u_1 = v_{x1}, u_2 = \omega_1, u_3 = \dot{\theta}$; $f(x) = \mathrm{M}^{-1}\begin{bmatrix} f_1 \\ f_2 \\ f_3 \end{bmatrix}$. The function $\mathrm{g}(\boldsymbol{x})$ is defined as $\mathrm{g}(\boldsymbol{x}) = \mathrm{M}^{-1}\begin{bmatrix} \frac{1}{R_1} & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} \mathrm{g}_{11} & \mathrm{g}_{12} & \mathrm{g}_{13} \\ \mathrm{g}_{21} & \mathrm{g}_{22} & \mathrm{g}_{23} \\ \mathrm{g}_{31} & \mathrm{g}_{32} & \mathrm{g}_{33} \end{bmatrix}$.

Therefore, $\mathrm{g}_1(\boldsymbol{x}) = \begin{bmatrix} \mathrm{g}_{11} \\ \mathrm{g}_{21} \\ \mathrm{g}_{31} \end{bmatrix}, \mathrm{g}_2(\boldsymbol{x}) = \begin{bmatrix} \mathrm{g}_{12} \\ \mathrm{g}_{22} \\ \mathrm{g}_{32} \end{bmatrix}, \mathrm{g}_3(\boldsymbol{x}) = \begin{bmatrix} \mathrm{g}_{13} \\ \mathrm{g}_{23} \\ \mathrm{g}_{33} \end{bmatrix}$. The functions $h(\boldsymbol{x})$ and $p(\boldsymbol{x})$ are defined as $h(\boldsymbol{x}) = \boldsymbol{x}$ and $p(\boldsymbol{x}) = \mathrm{M}^{-1}$. $\boldsymbol{w}$ is the system parameter uncertainty vector. The interference term is unpredictable and $p(\boldsymbol{x})\boldsymbol{w}$ is ignored in the decoupling process.

Definition 1 (Hassan, 2001) The conditions for the affine nonlinear system to be exactly linearizable are as follows:

(1) The matrix: $\left[\mathrm{g}_1(\boldsymbol{x}) \cdots \mathrm{g}_n(\boldsymbol{x}) L_f^{\gamma_1-1}\mathrm{g}_1(\boldsymbol{x}) \cdots L_f^{\gamma_n-1}\mathrm{g}_n(\boldsymbol{x})\right]$ is non-singular for all x belonging to the definite field $\mathbb{D}_0$;
(2) The vector field $\mathbb{D} = span\{\mathrm{g}_1(\boldsymbol{x}) \cdots \mathrm{g}_n(\boldsymbol{x})\}$ is collinear on $\mathbb{D}_0$.

If these conditions are satisfied, there exists a suitable set of output functions, such that the relative order vectors $[\gamma_1 \cdots \gamma_n]$ of the nonlinear system satisfy $\sum_{m=1}^{n} \gamma_m = n$, and the nonlinear system can be exactly linearized. Here, n represents the system matrix order. It should be noted that the control system is not enabled at $v_x = 0$, and when $\gamma_1 = 1$, then:

$$\begin{aligned} &L_f^0 h_1(x) = v_{x1}, L_f^0 h_2(x) = \omega_1, L_f^0 h_3(x) = \dot{\theta}, \\ &L_{g1}L_f^0 h_1(x) = \mathrm{g}_{11}, L_{g1}L_f^0 h_2(x) = \mathrm{g}_{12}, \\ &L_{g1}L_f^0 h_3(x) = \mathrm{g}_{13}, \\ &L_{g2}L_f^0 h_1(x) = \mathrm{g}_{21}, L_{g2}L_f^0 h_2(x) = \mathrm{g}_{22}, \\ &L_{g2}L_f^0 h_3(x) = \mathrm{g}_{23}, \\ &L_{g3}L_f^0 h_1(x) = \mathrm{g}_{31}, L_{g3}L_f^0 h_2(x) = \mathrm{g}_{32}, \\ &L_{g3}L_f^0 h_3(x) = \mathrm{g}_{33}. \end{aligned}$$

The relative orders $\lambda_1 = \lambda_2 = \lambda_3 = 1$ indicate that each state variable in the system has a relative degree of 1. In this case, the system matrix order $n = 3$ satisfies the condition $\lambda = \lambda_1 + \lambda_2 + \lambda_3 = 3 \le n$, which means that the system can be fully state feedback linearized. The decoupling matrix E, which facilitates the decoupling control, is given in Eq. (19)

$$\begin{aligned} \mathrm{E} = &\begin{bmatrix} L_{g1}L_f^{\lambda_1-1}h_1(x) & L_{g1}L_f^{\lambda_2-1}h_2(x) & L_{g1}L_f^{\lambda_3-1}h_3(x) \\ L_{g2}L_f^{\lambda_1-1}h_1(x) & L_{g2}L_f^{\lambda_2-1}h_2(x) & L_{g2}L_f^{\lambda_3-1}h_3(x) \\ L_{g3}L_f^{\lambda_1-1}h_1(x) & L_{g3}L_f^{\lambda_2-1}h_2(x) & L_{g3}L_f^{\lambda_3-1}h_3(x) \end{bmatrix} \\ = &\begin{bmatrix} \mathrm{g}_{11} & \mathrm{g}_{12} & \mathrm{g}_{13} \\ \mathrm{g}_{21} & \mathrm{g}_{22} & \mathrm{g}_{23} \\ \mathrm{g}_{31} & \mathrm{g}_{32} & \mathrm{g}_{33} \end{bmatrix} \neq 0. \end{aligned} \tag{19}$$

The non-singularity of the E matrix implies that a set of state variable substitutions, denoted as $z = \Phi(x)$, can be made to transform the system matrix into a standard regular type. In this case, the nonlinear variable substitutions are given by Eq. (20)

$$\boldsymbol{z} = \begin{bmatrix} L_f^{\lambda_1-1}h_1(x) \\ L_f^{\lambda_2-1}h_2(x) \\ L_f^{\lambda_3-1}h_3(x) \end{bmatrix} = \begin{bmatrix} v_{x1} \\ \omega_1 \\ \dot{\theta} \end{bmatrix}. \tag{20}$$

Setting a virtual input $\boldsymbol{v} = \begin{bmatrix} v_1 \\ v_2 \\ v_3 \end{bmatrix}$, the differential geometric decoupling control law is as in Eq. (21)

$$\begin{aligned} \boldsymbol{u} &= -\mathrm{E}^{-1}\left(\begin{bmatrix} L_f^{\lambda_1}h_1(\boldsymbol{x}) \\ L_f^{\lambda_2}h_2(\boldsymbol{x}) \\ L_f^{\lambda_3}h_3(\boldsymbol{x}) \end{bmatrix} + \begin{bmatrix} v_1 \\ v_2 \\ v_3 \end{bmatrix}\right) \\ &= -\mathrm{E}^{-1}\left(f(\boldsymbol{x}) + \begin{bmatrix} v_1 \\ v_2 \\ v_3 \end{bmatrix}\right). \end{aligned} \tag{21}$$

The new decoupled system is obtained after substitution into the original state array as shown in Eq. (22)

$$\begin{cases} \dot{\boldsymbol{z}} = \boldsymbol{v} + p(\boldsymbol{z})\boldsymbol{w} \\ \boldsymbol{y} = \boldsymbol{z} \end{cases}. \tag{22}$$

At this point, the new system has been decomposed into three single-input, single-output subsystems that do not interfere with each other, where v_1 is the tractor longitudinal velocity control command, v_2 is the tractor yaw rate control command, and v_3 is the trailer yaw rate control command. Since differential geometric decoupling cannot be controlled for disturbing factors with uncertainty, the $p(\boldsymbol{z})\boldsymbol{w}$ part that is ignored in the derivation of the differential geometric decoupling control law is controlled by the robust controller for anti-disturbance control.

3.2 Robust Controller Design

After the decoupling process, the new system is represented by Eq. (22). However, it should be noted that the disturbance term cannot be completely decoupled from the system during the decoupling process, as it remains uncertain. To address this issue, robust control techniques are employed to counteract the effects of external disturbances based on the linearized and decoupled system. Robust control provides a mechanism to enhance the system's stability and performance in the presence of uncertainties and disturbances, ensuring a more reliable and effective control strategy.

Assume the state reference as $z_{ref} = \begin{bmatrix} v_{x1\mathrm{ref}} \\ \omega_{1\mathrm{ref}} \\ \theta_{\mathrm{ref}} \end{bmatrix}$ and the system deviation as $\boldsymbol{e} = \boldsymbol{z} - \boldsymbol{z}_{\mathrm{ref}}$. Let $\omega_{1\mathrm{ref}} = \frac{v_{x1\mathrm{ref}}\delta}{(1+Kv_{x1\mathrm{ref}}^2)l_1}$, $\theta_{ref} = 0$, $v_{x1\mathrm{ref}}$ be given by the simulation condition. Then, the equation representing the system deviation can be expressed as follows:

$$\begin{cases} \dot{\boldsymbol{e}} = \boldsymbol{A}\boldsymbol{e} + \boldsymbol{B}_2\boldsymbol{v} + \boldsymbol{B}_1\boldsymbol{w} \\ \boldsymbol{z}_e = \boldsymbol{C}\boldsymbol{e} + \boldsymbol{D}_{12}\boldsymbol{v} + \boldsymbol{D}_{11}\boldsymbol{w} \\ \boldsymbol{y}_e = \boldsymbol{e} \end{cases}, \tag{23}$$

where $\boldsymbol{A} = \begin{bmatrix} 0&0&0 \\ 0&0&0 \\ 0&0&0 \end{bmatrix}$, $\boldsymbol{B}_2 = \mathrm{M}^{-1}$, $\boldsymbol{B}_1 = p(\boldsymbol{x})$, $\boldsymbol{C} = \begin{bmatrix} 1&0&0 \\ 0&1&0 \\ 0&0&1 \end{bmatrix}$, $\boldsymbol{D}_{12} = \boldsymbol{D}_{11} = \begin{bmatrix} 0&0&0 \\ 0&0&0 \\ 0&0&0 \end{bmatrix}$, $\boldsymbol{z}_e$ is the controlled error, and $\boldsymbol{y}_e$ is the output error.

Definition 2 (Lv & Wu, 2017) Consider the transfer function of the closed-loop system as $\boldsymbol{T}(\boldsymbol{s}) = \boldsymbol{D} + \boldsymbol{C}(\boldsymbol{sI} - \boldsymbol{A})^{-1}\boldsymbol{B}$. The following two statements are equivalent:

(1) The system is asymptotically stable, and the norm of $\boldsymbol{T}(\boldsymbol{s})$ is less than a given threshold, denoted as $\|\boldsymbol{T}(\boldsymbol{s})\| < \gamma$.
(2) There exists a positive definite symmetric matrix $\boldsymbol{X}$ satisfying the following matrix inequality:

$$\begin{vmatrix} \boldsymbol{A}^{\mathrm{T}}\boldsymbol{X} + \boldsymbol{X}\boldsymbol{A} & \boldsymbol{X}\boldsymbol{B} & \boldsymbol{C}^{\mathrm{T}} \\ \boldsymbol{B}^{\mathrm{T}}\boldsymbol{X} & -\gamma\boldsymbol{I} & \boldsymbol{D}^{\mathrm{T}} \\ \boldsymbol{C} & \boldsymbol{D} & -\gamma\boldsymbol{I} \end{vmatrix} < 0.$$

Theorem (Lv & Wu, 2017) *For the given controlled object described by Eq. (23), the existence of a state feedback H_∞ controller is determined by the satisfaction of a linear matrix inequality (LMI). Specifically, there exists a state feedback H_∞ controller, denoted as $\boldsymbol{v} = \left(\boldsymbol{K}\boldsymbol{X}^{-1}\right)\boldsymbol{e}$, if and only if there exists a symmetric positive definite matrix $\boldsymbol{X}$ and a matrix $\boldsymbol{K}$ that satisfy the LMI, as shown in Eq. (24)*

$$\begin{vmatrix} \boldsymbol{A}\boldsymbol{X} + \boldsymbol{B}_2\boldsymbol{K} + \left(\boldsymbol{A}\boldsymbol{X} + \boldsymbol{B}_2\boldsymbol{K}\right)^{\mathrm{T}} & \boldsymbol{B}_1 & \left(\boldsymbol{C}_1\boldsymbol{X} + \boldsymbol{D}_{12}\boldsymbol{K}\right)^{\mathrm{T}} \\ \boldsymbol{B}_1^{\mathrm{T}} & -\boldsymbol{I} & \boldsymbol{D}_{11}^{\mathrm{T}} \\ \boldsymbol{C}_1\boldsymbol{X} + \boldsymbol{D}_{12}\boldsymbol{K} & \boldsymbol{D}_{11} & -\gamma^2\boldsymbol{I} \end{vmatrix} < 0. \tag{24}$$

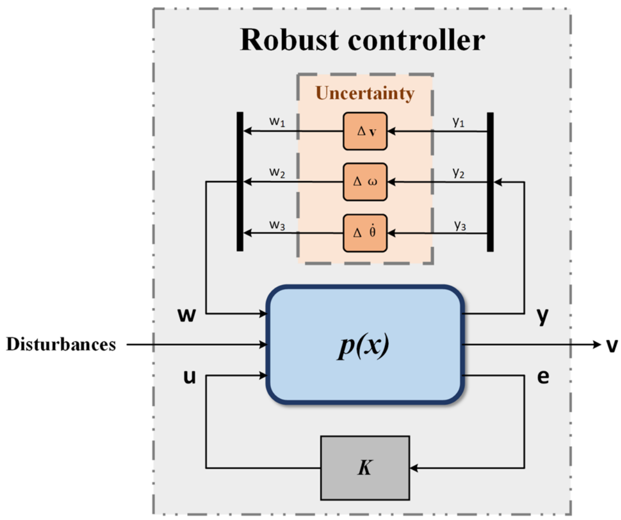

Fig. 3 Structure of the proposed robust controller

In the design of the robust controller, γ represents a definite bound on the H_∞ parametrization of the closed-loop system described by Eq. (12). The choice of γ plays a crucial role in determining the control effect. A smaller value of γ generally leads to better control performance. However, it is important to note that if the value of γ is chosen to be too small, it may result in the controller having no feasible solution (Wang et al., 2020). Therefore, there is a constraint on the selection of γ to ensure a balance between control effectiveness and feasibility

$$\begin{cases} trace\left(\boldsymbol{y}_e\right) < \rho \\ \sqrt{\rho} = \gamma_{\min} \end{cases}. \tag{25}$$

At this stage, the design of the robust controller is transformed into a standard linear LMI optimal solution problem. The structure of the proposed robust controller is shown in Fig. 3. This problem can be solved using the LMI Toolbox in MATLAB for Eqs. (24) and (25). By solving these LMIs, the final system control law can be obtained, as shown in Eq. (26)

$$\boldsymbol{u} = -\mathrm{E}^{-1}\left(f(\boldsymbol{x}) + \begin{bmatrix} v_1 \\ v_2 \\ v_3 \end{bmatrix}\right), \tag{26}$$

where $\begin{bmatrix} v_1 \\ v_2 \\ v_3 \end{bmatrix} = \left(\boldsymbol{K}\boldsymbol{X}^{-1}\right)\boldsymbol{e}$.

The control objectives of the proposed vehicle control system encompass longitudinal velocity tracking and stability control. Due to the high dependence of the motion decoupling on the vehicle model, the main parameters that can have an impact on the control performance are the main mechanism parameters of the AHV (as listed in Table 1) as well as the motion parameters (longitudinal velocity v_x, yaw rate ω, hinge angle $\theta, \ldots$) of the AHV obtained from sensor measurements or algorithmic estimations. The accuracy and real-time performance of obtaining these parameters also affect the control performance of this controller. The accuracy and real-time performance of parameter obtainment can be assured through the utilization of sensors characterized by high precision and a heightened sampling frequency. Meanwhile, the real-time performance of the proposed controller is predominantly associated with the control algorithm. In this study, the algorithm employed combines robust control and decoupling control based on low-order matrix operations. This approach yields the benefits of enhanced computational efficiency and superior timeliness, thereby enabling the proposed controller to meet the demands of real-time application.

Table 2 Parameters of the PI controller for stability control

Parameter	Value
K_p	2890
K_i	100

4 Model-in-the-Loop Simulation and Result Analysis

In this study, the effectiveness of the decoupling law and the performance of the robust controller in the vehicle control system are verified through three sets of comparative simulation tests. The simulations include different scenarios, and the results are compared among various vehicle configurations. The legend in the simulation results provides the necessary information for interpretation. "Reference" represents the reference or target values, while "Centralized Drive" refers to the vehicle with a centralized drive AHV as provided by TruckSim. "PI" represents a distributed electric drive AHV controlled by a PI without decoupling control. The PI controller parameters are set as shown in Table 2. "MPC" represents a distributed electric drive AHV controlled by an MPC without decoupling control. Finally, "Proposed Decoupling + Robust" corresponds to a distributed electric drive AHV equipped with the proposed decoupling controller and robust controller.

4.1 Single-Lane Change Simulation

During high-speed obstacle avoidance of the AHV, it is observed that the lateral motion of the trailer is often more

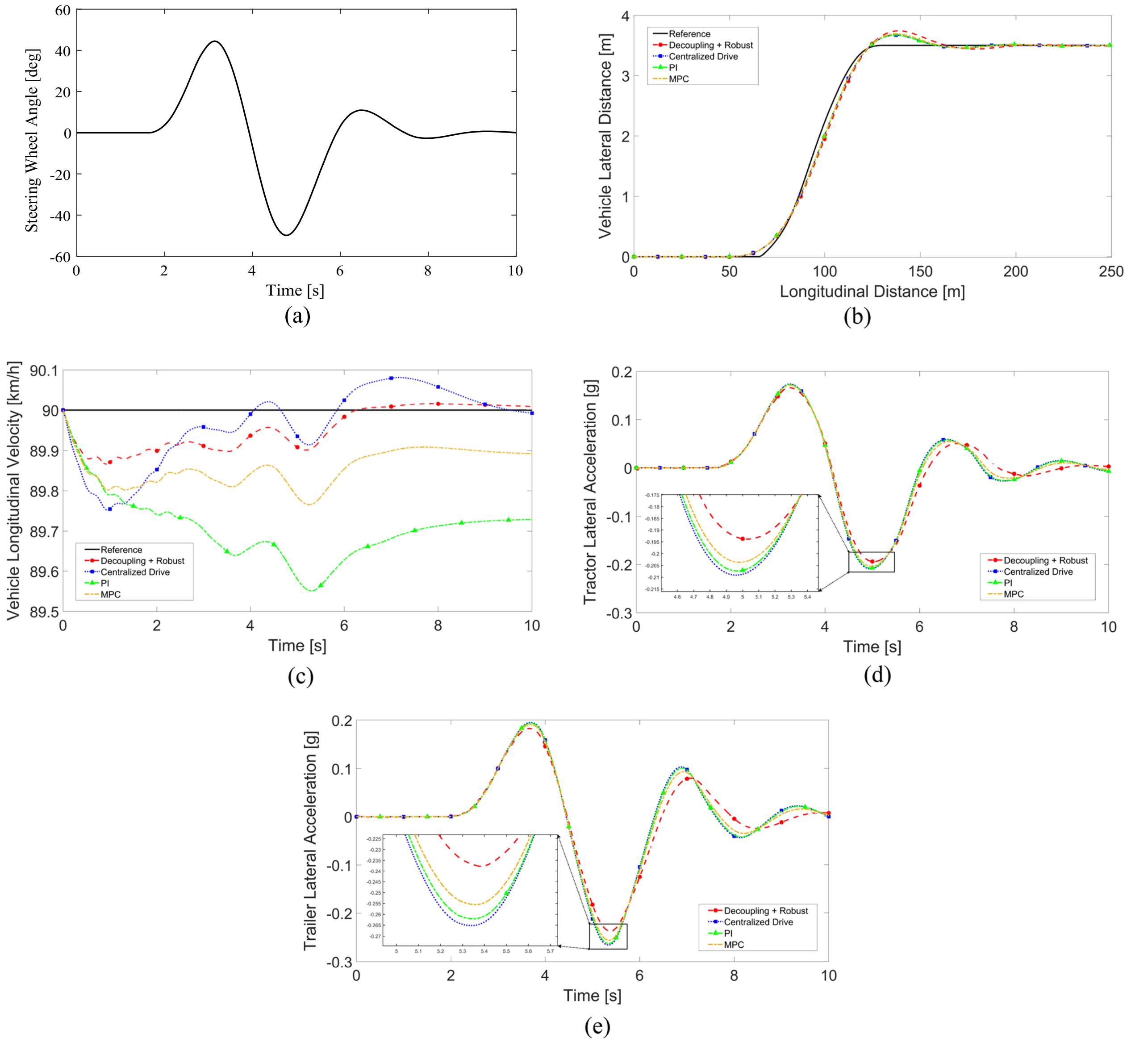

Fig. 4 Single line change simulation results: (**a**) steering wheel angle; (**b**) vehicle driving trajectory; (**c**) vehicle longitudinal velocity; (**d**) tractor lateral acceleration; (**e**) trailer lateral acceleration

pronounced compared to that of the tractor. The intensity of this phenomenon is measured by the rearward amplification (RA). The RA is defined as the ratio of the maximum lateral acceleration of the trailer unit to the maximum lateral acceleration of the tractor unit. The RA serves as a key parameter for evaluating the maneuvering stability performance of the AHV. This section establishes a single-lane change (SLC) test case with reference to the provisions of GB/T25979-2010 standard. A road surface adhesion coefficient of 0.85 was chosen for the simulations. The test scenario involved an SLC maneuver at a constant speed of 90 km/h. The graphical representation of the steering wheel angle input and the simulation results in Fig. 4 and the numerical data presented in Table 3 provide a comprehensive analysis of the AHV's lateral stability performance under these conditions.

Table 3 Maximum lateral acceleration of vehicles in SLC simulation

Simulation vehicle type	$\dot{v}_{y1}$ (g)	$\dot{v}_{y2}$ (g)	RA
Centralized drive	0.208	0.266	1.279
PI controlled	0.206	0.262	1.272
MPC controlled	0.203	0.257	1.266
Proposed decoupling + robust controlled	0.195	0.237	1.215

In Fig. 4b, it can be observed that the driving trajectories of the centralized drive vehicle, PI-controlled vehicle, MPC-controlled vehicle, and decoupling + robust controlled vehicle are all in close proximity to the reference trajectory. In Fig. 4c, the longitudinal velocity curve of the decoupling + robust controlled vehicle demonstrates less fluctuation and closely follows the target velocity, indicating its ability to maintain the reference vehicle velocity. Examining Fig. 4d, e, it is evident that the lateral acceleration of the trailer lags behind that of the tractor and enters a steady state, indicating the presence of RA. However, due to the tracking of the trailer yaw rate by the decoupling + robust controlled vehicle, the RA deviation is significantly reduced. Table 3 quantifies the reduction in the RA, with the decoupling + robust controlled vehicle exhibiting a 5.0% decrease compared to the centralized drive vehicle, a 4.5% decrease compared to the PI-controlled vehicle, and a 4.0% decrease compared to the MPC-controlled vehicle. These results highlight the improved stability of the decoupling + robust controlled vehicle under high-speed conditions.

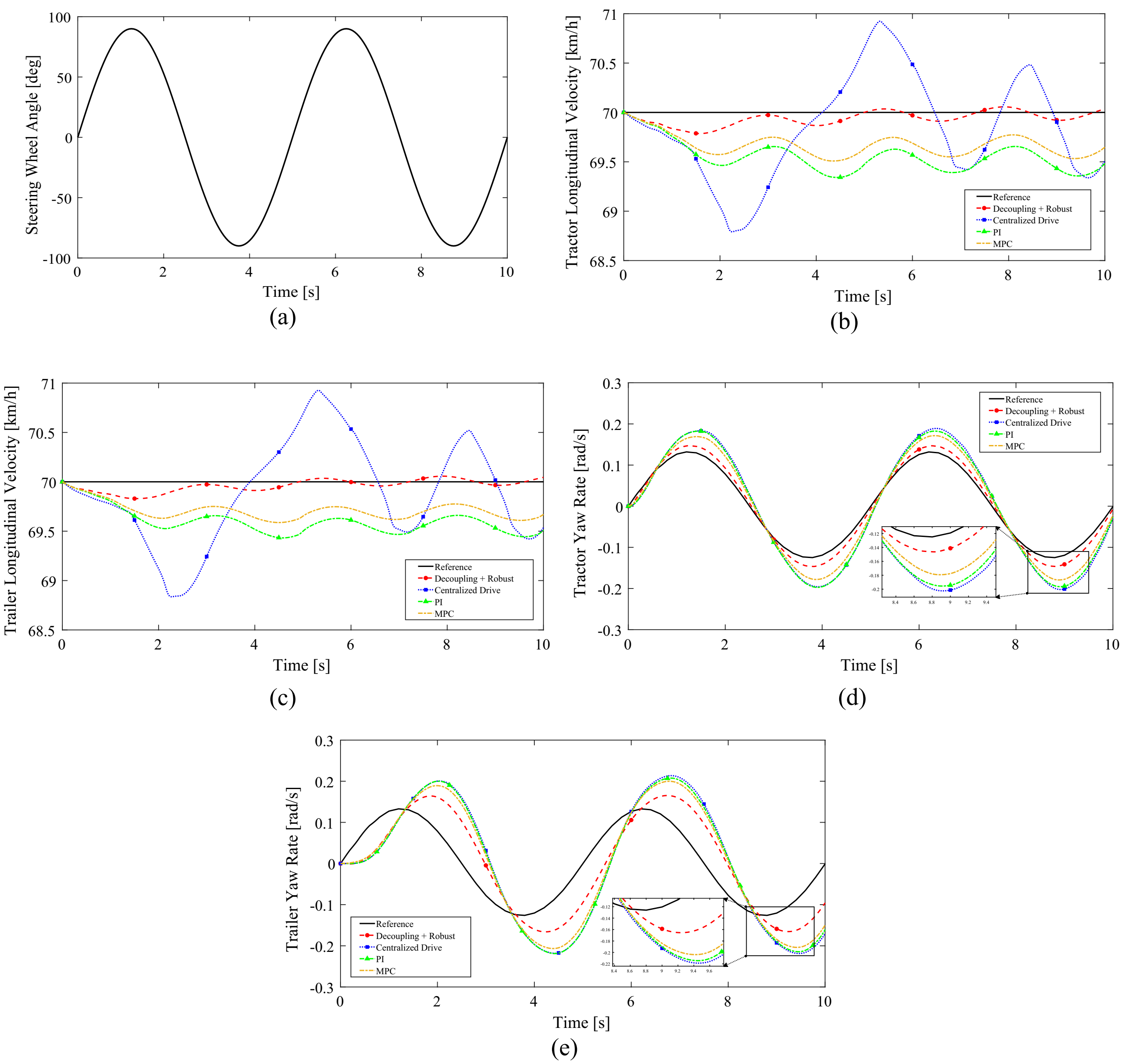

Fig. 5 Sinusoidal steering wheel angle input simulation results: (**a**) steering wheel angle; (**b**) tractor longitudinal velocity; (**c**) trailer longitudinal velocity; (**d**) tractor yaw rate; (**e**) trailer yaw rate

Table 4 Comparison of the maximum deviation of motion parameters

Simulation vehicle type and optimization effect	Δv_{x1} (km/h)	Δv_{x2} (km/h)	$\Delta\omega_1$ (rad/s)	j_2 (s)
Centralized drive	1.209	1.164	0.0775	0.658
PI controlled	0.658	0.568	0.0706	0.649
MPC controlled	0.492	0.413	0.0543	0.620
Proposed Decoupling + robust controlled	0.214	0.170	0.0214	0.439
Optimization rate of proposed method (compared to centralized drive)	82.3%	85.4%	72.4%	33.3%
Optimization rate of proposed method (compared to PI controlled)	67.5%	70.1%	69.7%	32.4%
Optimization rate of proposed method (compared to MPC controlled)	56.5%	58.8%	60.6%	29.2%

where j_2 represents the phase difference of the simulation yaw rate profile of the trailer compared to the reference yaw rate profile

4.2 Simulation of Sinusoidal Steering Wheel Angle Input

To validate the decoupling effect of the differential geometry decoupling control strategy in mitigating the coupling effect between yaw motion and longitudinal motion, a simulation test is conducted with a selected road surface adhesion coefficient of 0.85. The test encompasses a consistent velocity of 70 km/h, accompanied by a sinusoidal steering wheel angle input featuring an amplitude of $\pi/2$ and a frequency of $2\pi/5$ rad/s. The graphical representation of the steering wheel angle input and the corresponding simulation outcomes are exhibited in Fig. 5, while the numerical summary is provided in Table 4.

Figure 5b, c illustrates the influence of yaw motion on the vehicles. It can be observed that the decoupling + robust controlled vehicle exhibits the lowest frequency and smallest amplitude in the longitudinal velocity profile. The maximum deviation in the longitudinal velocity of the tractor is 0.214 km/h, which demonstrates an 82.3% reduction compared to the centralized drive vehicle, a 67.5% reduction compared to the PI-controlled vehicle, and a 56.5% reduction compared to the MPC-controlled vehicle. Additionally, the maximum deviation in the longitudinal velocity of the trailer is 0.17 km/h, reflecting a reduction of 85.4% compared to the centralized drive vehicle, 70.1% compared to the PI-controlled vehicle, and 58.8% compared to the MPC-controlled vehicle. The numerical results mentioned above is provided in Table 4. These results indicate that the longitudinal motion of the decoupling + robust controlled vehicle is minimally affected by the yaw motion. This outcome validates the achievement of a high degree of partial decoupling between the yaw motion and the longitudinal motion through the decoupling control strategy.

The reduction in longitudinal velocity deviation can be attributed to the decreased coupling between the yaw motion and the longitudinal motion. As a result, the error between the yaw rate of the decoupling-controlled vehicle's tractor and the reference yaw rate is minimized, as shown in Fig. 5d. Moreover, Fig. 5e and Table 4 demonstrate that the decoupling-controlled vehicle's trailer yaw rate exhibits both the smallest error and a reduced phase difference. This improvement indicates an enhanced tracking performance between the trailer and the tractor, as well as the suppression of the RA phenomenon, thanks to the effect of decoupling control.

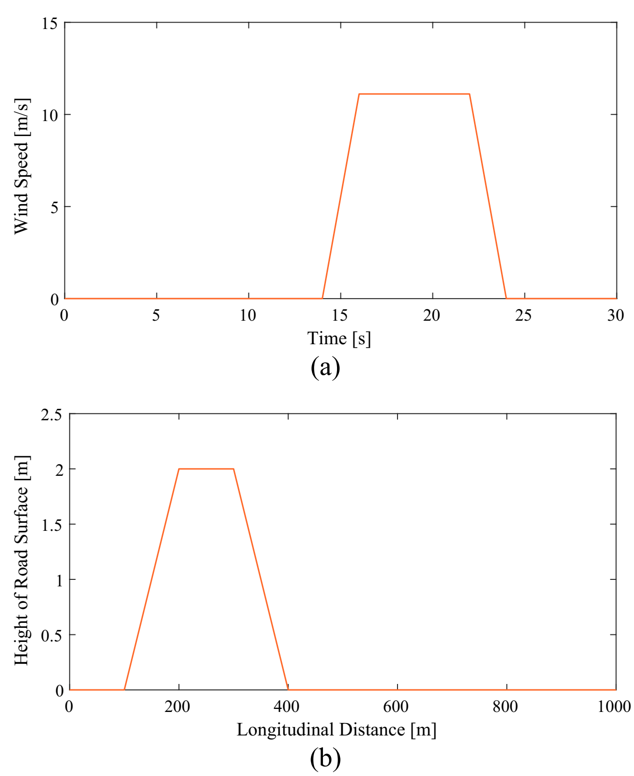

Fig. 6 Lateral wind and road height setting: (**a**) wind speed; (**b**) height of road surface

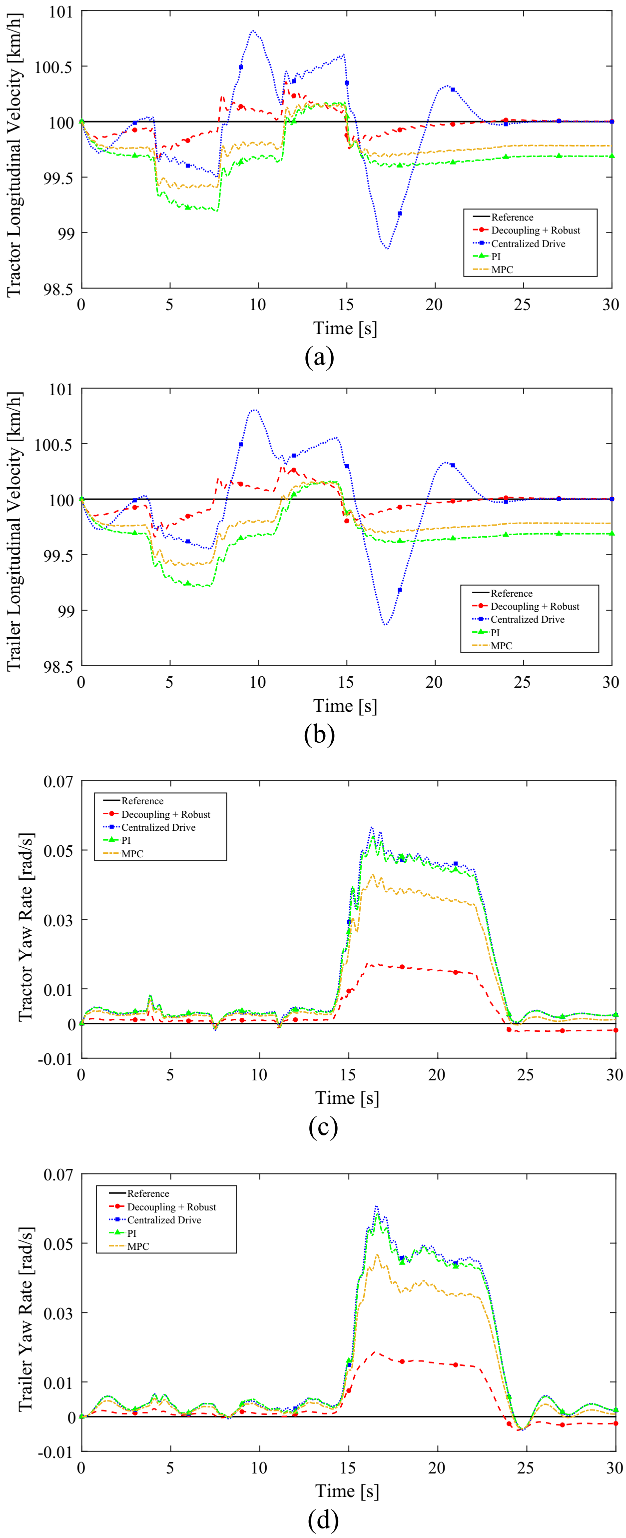

Fig. 7 Slope and lateral wind interference simulation results: (**a**) tractor longitudinal velocity; (**b**) trailer longitudinal velocity; (**c**) tractor yaw rate; (**d**) trailer yaw rate

4.3 Simulation of Slope and Lateral Wind Interference

The performance of the robust controller in handling longitudinal and lateral disturbances was evaluated through a simulation test. The test parameters included a constant speed of 100 km/h, a road adhesion coefficient of 0.85, a constant steering wheel angle input of 0°, and a constant wind direction of 90°. The wind conditions mentioned, with a wind speed of 11.11 m/s (approximately 40 km/h) and an intensity corresponding to level 6 (wind speed: 10.8 m/s ~ 13.8 m/s), fall into the category of strong wind. Strong winds can have a significant impact on vehicle dynamics and stability, particularly for high-profile vehicles such as AHVs. A straight-line test was conducted to assess the system's response to lateral wind and slope disturbances. The slope and lateral wind speed settings are shown in Fig. 6, and the simulation results are shown in Fig. 7.

Based on the observations from Fig. 7a, b, it is evident that the decoupling + robust controlled vehicle experiences the least impact on the longitudinal velocity of both the tractor and the trailer during the climbing stroke. This finding indicates that the decoupling + robust controlled vehicle demonstrates better anti-disturbance performance against longitudinal disturbances compared to the centralized drive vehicle, the PI-controlled vehicle, and the MPC-controlled vehicle. Furthermore, the presence of decoupling control results in reduced influence of lateral wind disturbances on the longitudinal motion of the AHV. As a result, the longitudinal velocity profile of the decoupling + robust controlled vehicle remains stable and closely follows the ideal velocity from the 14th to the 24th second.

By examining Fig. 7c, d, it becomes evident that the yaw rate of the decoupling + robust controlled vehicle demonstrates more stability and less deviation when subjected to lateral wind interference, in comparison to the PI-controlled vehicle, the MPC-controlled vehicle, and the centralized drive vehicle. Additionally, after the lateral wind ceases, the decoupling + robust controlled vehicle exhibits a smaller overshoot. These observations not only indicate that the longitudinal motion of the decoupling + robust controlled vehicle is less influenced by the lateral motion but also demonstrate the vehicle's improved resilience to lateral disturbances compared to the centralized drive vehicle, the PI-controlled vehicle, and the MPC-controlled vehicle. Moreover, the presence of decoupling control mitigates the effect of slope disturbances on the lateral motion of the AHV. As a result, the yaw rate profile of the decoupling + robust controlled vehicle remains less fluctuating until the 14th second.

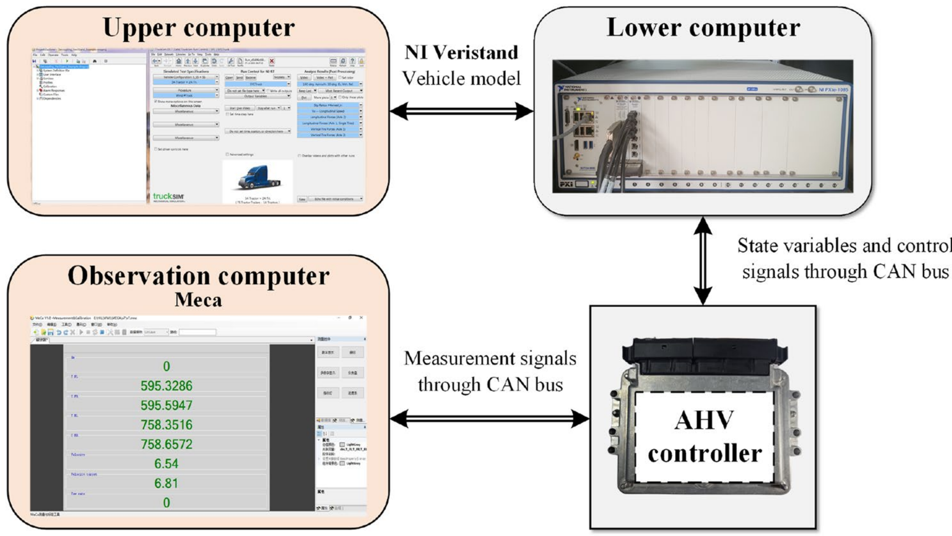

Fig. 8 Structure of the hardware-in-the-loop simulation platform

5 Hardware-in-the-Loop Experiment and Result Analysis

The hardware-in-the-loop simulation (HILs) platform consists of various hardware components, including an upper computer, a lower computer, and an AHV controller. The controller utilized in the platform is based on the MPC56xx series controller developed by the group. During the HILs' process, the upper computer configures the real-time test system using NI Veristand software. It then transmits the vehicle model and operating conditions to the lower computer via TruckSim. The control strategy model is compiled into a readable basic code for the controller using ECU Coder software. This code is then programmed into the controller using Meca software. The controller and lower computer are connected through CAN bus, enabling real-time communication and facilitating the execution of the simulation test. The structure of the HILs platform is shown in Fig. 8:

In the hardware-in-the-loop (HIL) experiments conducted to validate the control strategy's performance in real-time control, the researchers designed experiments using the HILs platform. To ensure consistency in the results, the three test conditions in the HILs test were kept the same as those in the model-in-the-loop simulation (MILs) test. In the legend of the experimental results, "Reference" refers to the reference value, "MILs" represents the results obtained from the model-in-the-loop simulation, and "HILs" represents the results obtained from the hardware-in-the-loop simulated experiment.

5.1 Single-Lane Change Experiment

The experimental simulation results are shown in Fig. 9 and Table 5.

In Fig. 9a, the trajectory curve is depicted, and the HILs results align closely with the MILs results. This indicates that the control strategy exhibits improved capability to follow the reference trajectory when implemented on the real controller. Figure 9b displays the longitudinal velocity curve, and the static difference observed in the HILs results during steady-state driving is smaller compared to the MILs results. Figure 9c, d illustrates the lateral acceleration curves of the tractor and trailer, respectively, showcasing that the HILs lateral acceleration response corresponds well to the MILs results.

It is important to note that, first, there exists a delay in the transmission of CAN bus communication signals within HILs. This delay, which is often overlooked in MILs, can potentially impact the performance of the control system. Second, due to the discrete-time solver used in HILs controllers, as opposed to the continuous-time solver employed in MILs controllers, there are inherent differences. These factors collectively contribute to a certain level of error between the longitudinal velocity of the HILs vehicle and the reference velocity during the early stages of the simulation (0.5 s ~ 2.1 s), as depicted in Fig. 9b, with a maximum error of up to 0.64 km/h. For a reference speed of 90 km/h, this error can be considered within a reasonable range. Additionally, noticeable oscillations are observed in all the result graphs within the first second of the simulation. However,

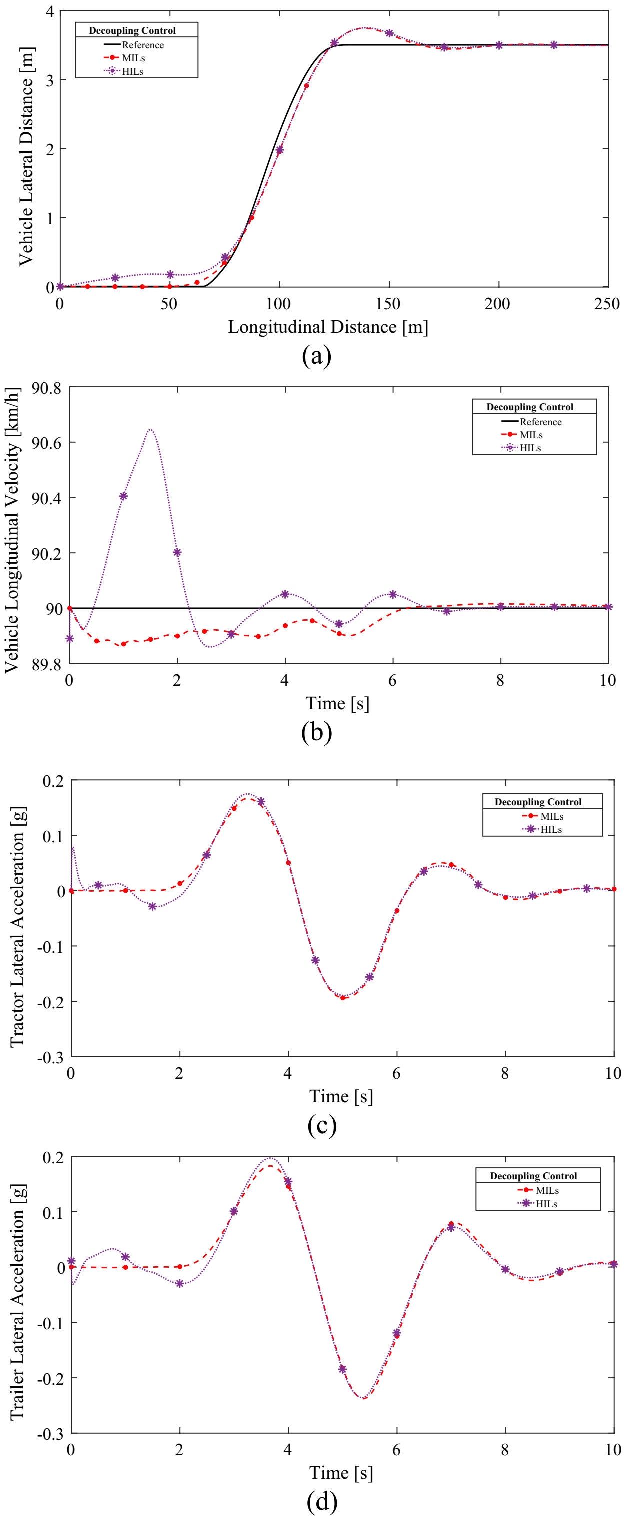

Fig. 9 Single line change experiment results: (**a**) vehicle driving trajectory; (**b**) vehicle longitudinal velocity; (**c**) tractor lateral acceleration; (**d**) trailer lateral acceleration

Table 5 Maximum lateral acceleration of vehicles for MILs and HILs

Simulation environment	$\dot{v}_{y1}$ (g)	$\dot{v}_{y2}$ (g)	RA
MILs	0.195	0.237	1.215
HILs	0.194	0.236	1.216

these oscillations quickly stabilize and reach a steady state within a relatively short period.

In Fig. 9a, the vehicle trajectory within the HILs, spanning from 0 to 50 m of longitudinal distance, demonstrates a marginal lateral deviation not exceeding 0.18 m from the driving trajectory observed in the MILs. In the HILs setup, the vehicle's longitudinal velocity undergoes a rapid surge from 0 to the target speed of 90 km/h, spanning the duration from the experiment's commencement to the conclusion of the initial solver step (0 ~ 0.0005 s). The swift augmentation in longitudinal velocity in conjunction with the previously noted oscillations in longitudinal velocity—most notably observed within the period of 0.24 s to 2.54 s—contributes to an initial instability in wheel torque output. This instability, in turn, engenders superfluous yaw moments. Consequently, the vehicle's trajectory within the HILs during the initial phase of the experiment deviates to a certain extent from the trajectory observed within the MILs. However, this deviation gradually diminishes as the vehicle initiates a lane change (at a longitudinal distance of 50 m). As the longitudinal distance progresses to 80 m, a more congruent alignment between the HILs and MILs trajectories is achieved (reducing the maximum lateral distance deviation between the 2 and 0.03 m).

5.2 Experiment of Sinusoidal Steering Wheel Angle Input

The results of MILs and HILs are shown in Fig. 10 and the comparison of vehicle motion parameters in steady state is shown in Table 6.

In Fig. 10a, b, the longitudinal velocity curves of the tractor and trailer are displayed. The HILs results exhibit similar errors under steady-state driving compared to the MILs results. This suggests that the control strategy remains effective in mitigating the coupling effect of yaw motion on longitudinal motion when implemented on the real controller. Figure 10c, d depicts the yaw rate curves of the tractor and trailer, respectively, demonstrating that the HILs yield yaw rate responses consistent with the MILs results.

Due to the utilization of a real controller and communication via the CAN bus in the HILs test, some fluctuations occur at the initial phase of the test. As a result, there

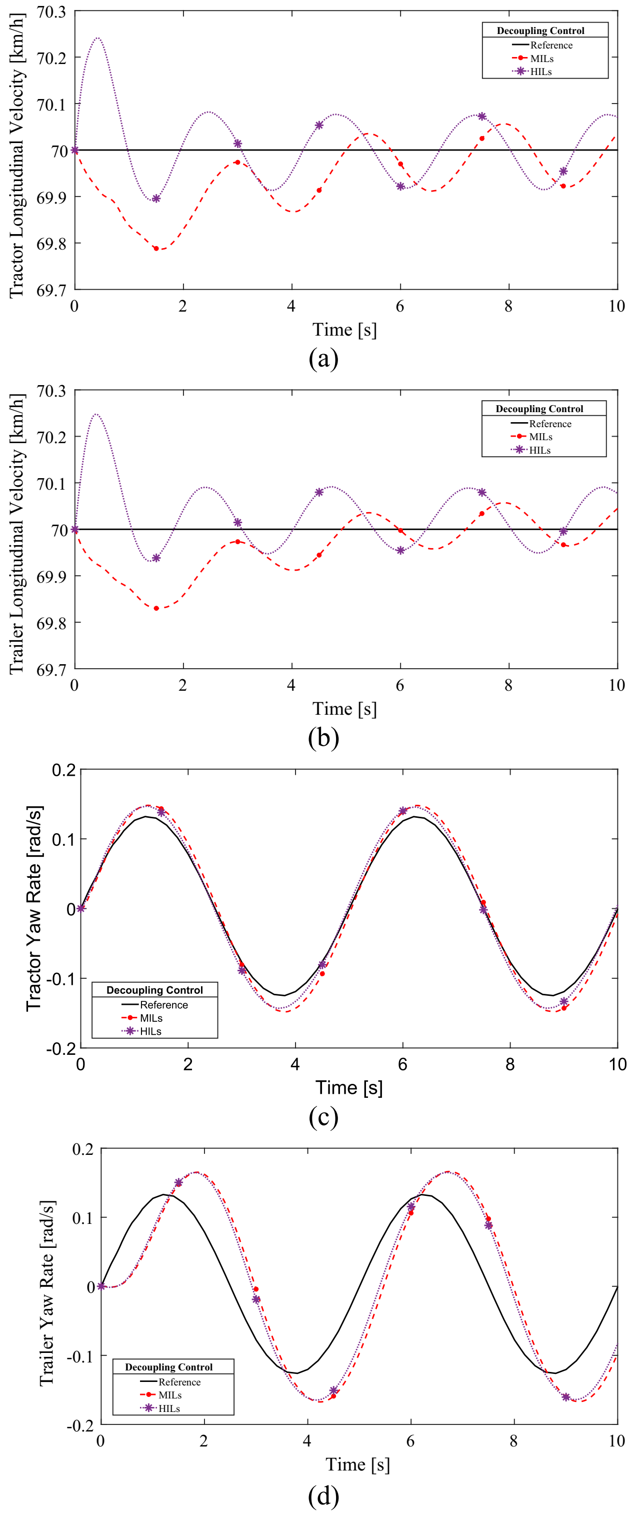

Fig. 10 Sinusoidal steering wheel angle input experiment results: (**a**) tractor longitudinal velocity; (**b**) trailer longitudinal velocity; (**c**) tractor yaw rate; (**d**) trailer yaw rate

Table 6 Extremal deviation of motion parameters for MILs and HILs in steady state

Simulation environment	Δv_{x1} (km/h)	Δv_{x2} (km/h)	$\Delta \omega_1$ (rad/s)	j_2 (s)
MILs	0.072	0.042	0.0214	0.439
HILs	0.082	0.046	0.0210	0.431

exists a phase difference between the MILs and the HILs in Fig. 10a, b.

5.3 Experiment of Slope and Lateral Wind Interference

The results of the experimental simulation are shown in Fig. 11.

Figure 11a, b demonstrates that the longitudinal velocity tracking is improved and less influenced by slope disturbances, indicating that the control strategy exhibits better immunity against external longitudinal disturbances in the real controller. However, due to the oscillation of the yaw rate shown in Fig. 11c, d during the simulation period from 0 to 3 s, there exists a significant difference between the HILs results of the yaw rate and the MILs results. Nevertheless, this oscillation allows for a quicker recovery to a steady state. Additionally, under lateral wind disturbances, the HILs yaw rate exhibits smaller fluctuations, suggesting that the control strategy demonstrates better anti-interference capability against external lateral disturbances in the real controller.

6 Conclusion

In this paper, a nonlinear three-degree-of-freedom vehicle model is utilized to address the motion control problem of a distributed electric drive AHV. A decoupling controller for the AHV is designed based on the differential geometry method, effectively reducing the coupling influence between the longitudinal and lateral motions of the vehicle. Additionally, considering the impact of external wind disturbances during actual driving, an anti-disturbance robust controller is developed. Simulation analysis and HILs tests are conducted to evaluate the performance of the proposed control strategies. The results demonstrate significant improvements compared to an AHV without decoupling control. Specifically, the RA is reduced by 4.5%, the longitudinal velocity deviation is reduced by 67.5%, and the yaw rate deviation is reduced by 69.7% against the PI-controlled distributed drive AHV. Moreover, the vehicle exhibits enhanced anti-disturbance performance against external disturbances.

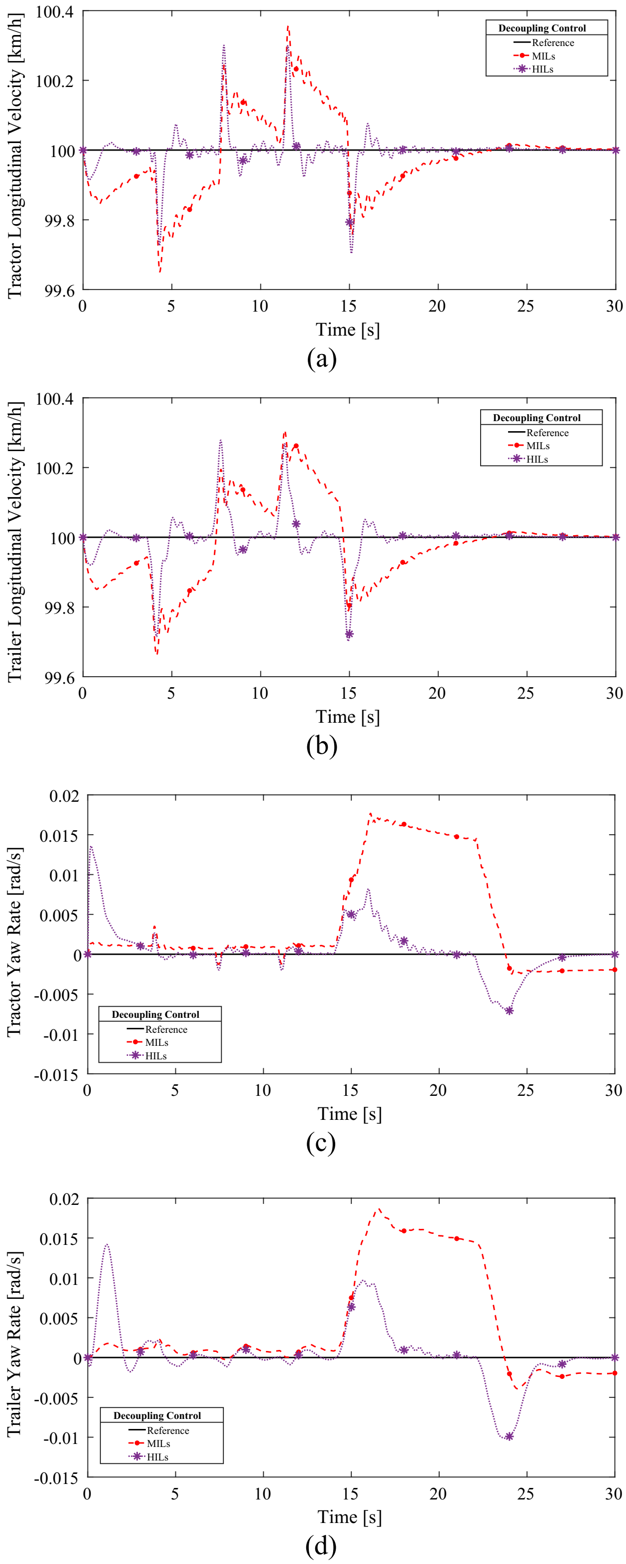

Fig. 11 Slope and lateral wind interference experiment results: (**a**) tractor longitudinal velocity; (**b**) trailer longitudinal velocity; (**c**) tractor yaw rate; (**d**) trailer yaw rate

Meanwhile, the real-time application of the designed controller is verified to be effective through HILs.

This paper conducts specific research concerning the decoupling of dynamics in the distributed drive AHV system. During the design and simulation of the control strategy, we have traditionally operated under the assumption that the ground imparts an ample reactive force onto the tires. We have thus far disregarded the potential impact of wheel slip or instances where the wheels lose contact with the ground on the efficacy of control. In the future, making the control strategy more compatible with realistic driving conditions by thoroughly investigating the effect of wheel slip on the dynamics coupling can be taken as a further research topic.

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Data availability This data relates to a project that has not yet been completed, and the data that support the findings of this study are not publicly available at this time.

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